

RESEARCH PAPER ANALYSIS

TEAM – Quantum CodeX

1. **Title:** Characterizing Mental Health and Wellness in Students Across Engineering Disciplines (2018)

Merits:

1. **Comprehensive Coverage:**
The study analyzes multiple mental health conditions (depression, anxiety, PTSD, substance abuse) across **different engineering disciplines**, giving a broad and realistic picture.
2. **Use of Validated Instruments:**
Standard and clinically validated tools like **PHQ**, **Kessler-6**, **CAGE-AID**, and **PC-PTSD** ensure reliability and credibility of results

Demerits:

1. **Single-Institution Study:**
Data was collected only from **California Polytechnic State University**, limiting generalizability to other universities.
2. **Self-Reported Data Bias:**
Responses depend on students' self-reporting, which may include exaggeration, under-reporting, or selective participation.

Objecti

1. To **compare mental wellness across different engineering disciplines**.
2. To **identify high-risk student groups** needing mental health support.

Limitations:

- Low response rate (~16%) may skew results toward students already experiencing distress
- No causal relationship established between engineering workload and mental illness
- Lack of qualitative interviews to explain *why* issues occur
- Absence of comparison with non-engineering students

Model / Tools Used:

The study does **not use machine learning models**.

Instead, it uses **psychological screening models**, including:

- **PHQ (Patient Health Questionnaire)** – depression & anxiety
- **Kessler-6 (K6)** – serious mental illness
- **CAGE-AID** – substance abuse
- **PC-PTSD** – post-traumatic stress disorder

Methodology (How Objectives Were Fulfilled):

1. **Survey Design:**
Combined multiple validated mental health questionnaires into one online survey.
2. **Data Collection:**
Survey distributed via email and flyers to engineering students across 9+ disciplines.
3. **Data Cleaning & Analysis:**
Invalid responses removed; results analyzed discipline-wise to identify trends.
4. **Ethical Compliance:**
Data stored under HIPAA-compliant security with no personal identifiers.

Future Scope:

1. Conduct **multi-institution and nationwide studies** for broader insights
2. Include **longitudinal tracking** to observe mental health changes over time
3. Integrate **AI/ML-based prediction models** for early risk detection
4. Develop **discipline-specific mental wellness interventions**
5. Compare engineering students with **non-engineering populations**

2. Title: *Generative AI–Based Mental Wellness Chatbot(2025)*

Merits :

1. **24/7 Mental Health Support:**
The chatbot provides continuous, on-demand emotional support, making mental wellness assistance accessible anytime without dependency on human availability.
2. **Personalized Interaction Using GenAI:**
By using Generative AI and NLP, the chatbot adapts responses based on user input, offering more personalized and empathetic conversations compared to rule-based chatbots.

Demerits:

1. **Lack of Human Emotional Depth:**
Despite advanced AI, the chatbot cannot fully replicate human empathy, emotional understanding, or clinical judgment.
2. **Risk of Over-Reliance:**
Users may rely heavily on the chatbot instead of seeking professional mental health care when needed.

Objectives :

1. To **develop an AI-based chatbot** capable of providing mental wellness support through natural conversation.
2. To **analyze user sentiment and emotional state** using text-based interactions.
3. To **offer coping strategies and mental health guidance** in a non-judgmental environment.

Limitations:

- Chatbot responses depend heavily on training data quality
- Not suitable for handling severe mental health crises
- Ethical concerns regarding data privacy and emotional safety
- Limited understanding of cultural and contextual nuances

Model Used:

The system uses **Generative AI with NLP models**, including:

- **Transformer-based language models**
- **Natural Language Processing (NLP)** for intent and sentiment detection
- **Rule-based fallback logic** for safety-critical responses

These models enable contextual understanding and response generation.

Methodology (How Objectives Were Fulfilled):

1. **Data Collection:**
Mental health–related conversational datasets were used to train the chatbot.
2. **Model Training:**
Generative AI models were trained to understand emotional cues and generate empathetic responses.
3. **Sentiment Analysis:**
NLP techniques were applied to classify user input into emotional categories such as stress, anxiety, or positivity.
4. **Response Generation:**
Based on detected sentiment, appropriate mental wellness suggestions and supportive dialogue were generated.
5. **Evaluation:**
User interaction quality and response relevance were evaluated through testing and feedback.

Future Scope:

1. Integration with **real-time professional support systems**
2. Use of **multimodal inputs** (voice, facial expression analysis)
3. Incorporation of **crisis-detection and emergency escalation**
4. Continuous learning from user feedback
5. Deployment as a **mobile and wearable-integrated mental wellness assistant**

3. Synthesizing AI for Mental Wellness(2024)

Merits:

1. **24/7 Accessible Mental Health Support**
AI-based systems provide continuous, on-demand mental wellness assistance without time or location constraints.
2. **Personalized and Scalable Care**
AI can tailor responses using user inputs, emotional patterns, and behavioral data, making support scalable for large populations.

Demerits:

1. **Lack of Human Emotional Depth**
AI cannot fully replicate human empathy, emotional intuition, or therapeutic judgment.
2. **Risk of Misinterpretation**
Incorrect sentiment or context detection may lead to unsuitable or ineffective responses.

Objectives:

1. To **integrate AI technologies** for monitoring and supporting mental wellness.
2. To **analyze user emotional and psychological patterns** using AI-driven models.
3. To **provide early mental health intervention** and reduce dependency on traditional in-person services.

Limitations:

- Dependency on **self-reported user data**, which may be biased or incomplete
- **Not suitable for crisis situations** or severe mental health disorders
- Data privacy and ethical concerns related to sensitive mental health information

Model Used:

- **Generative AI / NLP Models**
 - Large Language Models (LLMs)
 - Sentiment Analysis Models
- **Machine Learning Techniques**
 - Emotion classification
 - Behavioral pattern recognition
- **Therapeutic Framework Integration**
 - Cognitive Behavioral Therapy (CBT)
 - Mindfulness-based approaches

Methodology to Fulfill Objectives:

1. **Data Collection**
 - a. User text inputs (chat conversations)
 - b. Optional voice inputs (tone, stress indicators)
2. **Preprocessing**
 - a. Text normalization
 - b. Emotion and sentiment tagging
3. **AI Analysis**
 - a. NLP models detect emotional states (stress, anxiety, positivity)
 - b. Context-aware response generation
4. **Wellness Assessment**
 - a. Mental wellness scoring
 - b. Indicator detection (anxiety, depression, engagement)
5. **Feedback & Support Delivery**
 - a. Personalized coping strategies
 - b. Psychoeducational content and recommendations

Future Scope:

- 1. **Multimodal Analysis**
 - a. Combine text, voice, facial expressions, and physiological data
- 2. **Clinical Integration**
 - a. AI-assisted tools for therapists and counselors
- 3. **Real-Time Crisis Detection**
 - a. Emergency alerts and escalation mechanisms
- 4. **Improved Ethical AI Frameworks**
 - a. Stronger data privacy, bias reduction, and explainability

4. Title: AI-Based and Digital Mental Health Apps: Balancing Need and Risk (2023)

Merits:

- 1. **Improved Accessibility to Mental Health Care**
AI-based and digital mental health apps provide affordable, scalable, and 24/7 access to mental health support, especially for underserved and remote populations.
- 2. **Early Detection and Self-Management**
These apps help users monitor symptoms, track mood, and detect early signs of mental distress, enabling timely self-intervention and awareness.

Demerits :

- 1. **Data Privacy and Security Risks**
Mental health apps collect sensitive personal data, raising concerns about data misuse, breaches, and lack of transparent data governance.
- 2. **Limited Clinical Validation**
Many apps lack strong clinical evidence or regulatory approval, making their effectiveness and safety uncertain.

Objectives:

- 1. To **analyze the growing demand** for AI-based and digital mental health applications.
- 2. To **evaluate benefits and potential risks** associated with these applications.
- 3. To **highlight ethical, regulatory, and safety challenges** in deploying mental health technologies.

Limitations:

- Over-reliance on self-reported user data
- Lack of standardized evaluation frameworks
- Insufficient long-term effectiveness studies
- Inconsistent regulation across countries

Model / Technologies Used:

- **AI & Machine Learning Models**
 - Natural Language Processing (NLP)
 - Sentiment and emotion analysis
- **Digital Health Frameworks**
 - Mood tracking systems
 - Cognitive Behavioral Therapy (CBT)–based modules
- **Rule-based and hybrid AI systems** for mental health guidance

Methodology (How Objectives Were Fulfilled):

1.	Literature	Review
	Analysis of existing AI-based mental health apps, clinical studies, and regulatory guidelines.	
2.	Risk-Benefit	Assessment
	Evaluation of usability, accessibility, effectiveness, and potential harms of digital mental health solutions.	
3.	Ethical and Regulatory	Analysis
	Examination of privacy policies, consent mechanisms, and compliance with health regulations.	
4.	Comparative	Analysis
	Comparison between traditional mental health care and AI-based digital interventions.	

Future Scope:

1.	Development of strong regulatory frameworks for mental health apps
2.	Improved clinical validation and evidence-based design
3.	Integration of AI systems with professional mental health services
4.	Enhanced data privacy, transparency, and ethical AI practices
5.	Adoption of personalized and culturally adaptive AI models

5. Title: AI Used to Tackle Mental Health Issues (2024)

Merits:

1.	Improved Access to Mental Health Support
	AI-based tools such as chatbots, mood trackers, and virtual therapists make mental health support easily accessible, especially for people in remote or underserved areas.
2.	Early Detection and Continuous Monitoring
	AI systems can analyze user behavior, speech, and text patterns to detect early signs of mental health issues like anxiety and depression, enabling timely intervention.

Demerits:

1.	Limited Emotional Understanding
	AI lacks genuine human empathy and may not fully understand complex emotional or psychological conditions.
2.	Privacy and Ethical Concerns
	The collection and processing of sensitive mental health data raise concerns about data security, consent, and ethical usage.

Objectives:

1.	To explore the role of AI technologies in addressing mental health challenges.
2.	To analyze how AI can assist in diagnosis, monitoring, and therapy for mental health issues.
3.	To evaluate the benefits and challenges of using AI in mental health care.

Limitations:

•	Dependence on self-reported and digital data
•	Not suitable for handling severe psychiatric emergencies
•	Risk of algorithmic bias affecting diagnosis

- Lack of universal clinical validation

Model / Technologies Used:

- **Artificial Intelligence & Machine Learning Models**
 - Natural Language Processing (NLP)
 - Sentiment and emotion analysis
- **Chatbots and Virtual Assistants**
- **Predictive Analytics Models**
- **Cognitive Behavioral Therapy (CBT)–based AI systems**

Methodology (How Objectives Were Fulfilled):

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|--|--|
| <ol style="list-style-type: none"> 1. Data
User interaction data, text inputs, and behavioral patterns are collected through digital platforms. 2. Data
Cleaning, normalization, and labeling of data for emotion and sentiment analysis. 3. AI Analysis
Machine learning and NLP models analyze emotional states and mental health indicators. 4. Intervention & Support
Based on analysis, AI systems provide coping strategies, alerts, or recommendations for professional help. | <p>Collection</p> <p>Preprocessing</p> |
|--|--|

Future Scope:

1. Integration of **multimodal inputs** (voice, facial expression, wearable data)
2. Stronger **ethical and regulatory frameworks**
3. AI-assisted tools for **mental health professionals**
4. Improved personalization through **adaptive learning models**
5. Global deployment of affordable AI mental health solutions

6. First-Year College Students: Perspectives on Technology and Wellness in Education (2025)

Merits:

1. **Holistic View of Student Wellness**
The paper connects mental health, academic pressure, technology use, and life transitions, giving a well-rounded understanding of first-year students' wellness challenges.
2. **Timely and Relevant Context**
It strongly addresses post-COVID impacts, virtual schooling effects, AI tools (like ChatGPT), and digital over-reliance, making it highly relevant to current education systems.

Demerits:

1. **Lack of Quantitative Data**
The paper is largely conceptual and literature-based, with limited original statistical or experimental validation.
2. **Region-Specific Perspective**
Much discussion is centered on Canadian education systems, which may reduce direct applicability to other countries without adaptation.

Objectives of the Study:

1. To analyze how **technology usage affects first-year students' mental wellness**.
2. To explore the **challenges faced during transition from high school to university**.
3. To propose **ways institutions can balance technology integration with wellness practices**.

Methodology Used:

- **Literature Review Approach**
The author reviews existing research, theories, surveys, and prior studies related to:
 - Student mental health
 - Digital wellness
 - AI and education
 - Academic writing challenges
- **Conceptual Analysis**
Uses wellness models (e.g., Hettler's Six Dimensions of Wellness) to frame discussions.

Model Used:

- **Hettler's Six Dimensions of Wellness Model**
 - Emotional
 - Intellectual
 - Physical
 - Social
 - Occupational
 - Spiritual

Limitations of the Study:

1. No primary experimental or survey data collected by the author.
2. Focuses mainly on first-year university students, excluding later academic years.
3. Rapid technological changes (AI tools) may outdate some observations quickly.

Future Scope:

1. Conduct **empirical studies** involving surveys or experiments with first-year students.
2. Explore **AI-assisted learning tools** designed specifically for wellness support.
3. Compare **cross-cultural student wellness** across different countries.
4. Develop **technology-wellness policies** for universities.

How the Objectives Are Fulfilled:

- By reviewing **post-pandemic studies**, the paper explains current stressors.
- By applying **wellness frameworks**, it connects technology use with mental health.
- By suggesting **curriculum reforms and digital wellness practices**, it provides practical institutional recommendations.

7. Impact of Artificial Intelligence Chatbots on Student Well-being and Mental Health (2021)

Merits of the Study:

1. The study provides a **comprehensive overview** of how AI chatbots influence student mental health across schools and universities.
2. It highlights **practical applications** of chatbots in counseling, advising, and learning, offering useful insights for educators and policymakers.

Demerits of the Study:

1. The study is limited to **a small number of selected articles**, which may restrict the generalizability of the findings.
2. Most reviewed studies focus on **university students**, giving less attention to high school students.

Limitations of the Study:

1. The literature review was restricted to **specific databases**, excluding major sources such as Scopus and Web of Science.
2. The search was limited to **article titles only**, which may have resulted in missing relevant studies found in abstracts or keywords.

Objective of the Study:

1. To examine the impact of AI-based chatbots on students' mental health and psychological well-being.
2. To identify how educational institutions use chatbots for counseling, advising, and learning support to reduce stress, anxiety, and depression among students.

Methodology to Fulfill the Objective:

1. A **Systematic Literature Review (SLR)** was conducted using the PRISMA framework to identify, screen, and select relevant studies published between 2019 and 2023.
2. Multiple academic databases were searched using predefined keywords related to chatbots, mental health, and education, followed by inclusion and exclusion criteria to finalize eligible studies.

Future Scope of the Research:

1. Future research can include **additional databases and broader search strategies** to capture more relevant studies.
2. Further studies can focus on **advanced chatbot techniques**, emotional intelligence, and long-term psychological effects on students across different educational levels.

8. Artificial Intelligence and Social Media on Academic Performance and Mental Well-being(2024)

Merits of the Study:

1. The study provides **empirical evidence** showing the positive role of AI and social media in improving both academic performance and mental well-being.
2. The inclusion of **smart learning as a mediator** offers a strong theoretical and practical contribution to technology-enhanced education research.

Demerits of the Study:

1. The research relies on **self-reported data**, which may introduce response bias or subjectivity.

2. The study focuses only on **Chinese university students**, limiting the generalizability of results to other countries or educational contexts

Limitations of the Study:

1. The research design is **cross-sectional**, making it difficult to establish long-term or causal relationships.
2. The study mainly considers **Generative AI (e.g., ChatGPT)** and does not compare multiple AI technologies or platforms.

Objective of the Study:

1. To examine students' perceptions of how **artificial intelligence and social media** influence academic performance and mental well-being.
2. To analyze the **mediating role of smart learning** in strengthening the relationship between AI, social media, academic performance, and mental well-being.

Methodology to Fulfill the Objective:

1. A **quantitative, cross-sectional survey method** was adopted, collecting data from **401 university students** using structured questionnaires and a five-point Likert scale.
2. **Partial Least Squares-Structural Equation Modeling (PLS-SEM)** was used via **SmartPLS software** to test hypotheses, measure relationships, and analyze mediation effects.

Model Used in the Study:

1. The study employed a **Conceptual Structural Model based on Social Learning Theory (SLT)**, where **AI and Social Media** act as independent variables.
2. **Smart Learning** functions as a **mediating variable**, while **Academic Performance** and **Mental Well-being** are dependent variables, analyzed using **PLS-SEM**.

Future Scope of the Research:

1. Future studies can adopt a **longitudinal or experimental design** to better understand causal effects over time.
2. Research can be expanded to include **diverse countries, age groups, and additional AI models**, such as machine learning analytics and adaptive tutoring systems.

9. AI-Enabled Smartphone-Based Intervention Mental Health Application for University Students (2025)

Merits of the Study:

1. The study proposes a **comprehensive AI-driven mental health solution** integrating sentiment analysis, biofeedback, peer support, and chatbots within a single smartphone application.
2. It emphasizes a **preventive and early-intervention approach**, making mental health support more accessible, scalable, and cost-effective for university students.

Demerits of the Study:

1. The study primarily focuses on **conceptual framework and design**, with limited real-world deployment and empirical evaluation.

2. The effectiveness of the app depends heavily on **user engagement and data privacy management**, which may vary across individuals.

Limitations of the Study:

1. The proposed system lacks **large-scale experimental validation**, as the app was not fully tested across diverse student populations.
2. The research relies on **self-reported data and expert feedback**, which may introduce subjective bias.

Objective of the Study:

1. To **design and propose an AI-enabled smartphone-based mental health application** that supports university students' emotional, social, and psychological well-being.
2. To utilize **artificial intelligence, biofeedback, and peer support mechanisms** for early intervention, loneliness reduction, and mental wellness improvement among students.

Methodology to Fulfill the Objective:

1. A **design-oriented research approach** was adopted using scoping research, expert interviews, and user feedback to develop the mental health application framework.
2. **AI techniques such as sentiment analysis, biofeedback monitoring, and chatbot interaction** were integrated into the app design to assess user emotions and provide personalized mental health support.

Model Used in the Study:

1. The study employed a **multi-theoretical conceptual model** combining **Nudge Theory, Social Support Theory, Belongingness Theory, Social Presence Theory, Theory of Planned Behavior, Social Information Processing Theory, and Sociocultural Theory**.
2. These theories collectively guide the **mHealth app framework**, linking AI-driven interventions, social interaction, and psychological well-being outcomes for university students.

Future Scope of the Research:

1. Future work can involve **pilot testing and large-scale deployment** of the application to evaluate usability, effectiveness, and long-term mental health outcomes.
2. Advanced **machine learning models, wearable sensor integration, and personalized AI interventions** can be further enhanced to improve accuracy and user experience.

10. Emotional Intelligence Meets Artificial Intelligence: Affective Computing and Student Well-Being in Online Classrooms (2025)

Merits of the Study:

1. The research provides **strong empirical evidence** linking emotional intelligence and affective computing with improved student well-being and academic engagement.
2. It highlights the **practical value of emotion-aware AI tools**, such as sentiment analysis and intelligent feedback systems, in reducing dropout rates.

Demerits of the Study:

1. The study relies on **self-reported questionnaire data**, which may be affected by response bias.
2. The research is **context-specific to Pakistani universities**, limiting generalizability to other educational systems.

Limitations of the Study:

1. The **cross-sectional research design** restricts the ability to analyze long-term effects of emotionally intelligent AI systems.
2. Ethical concerns such as **privacy, emotional surveillance, and algorithmic bias** were discussed but not empirically tested.

Objective of the Study:

1. To examine the **role of emotional intelligence (EI)** in enhancing students' academic well-being and engagement in online learning environments.
2. To analyze how **affective computing-based AI systems** predict emotional states, academic engagement, and dropout risks in virtual classrooms.

Methodology to Fulfill the Objective:

1. A **quantitative, correlational, and predictive research design** was employed using data collected from **250 university students** enrolled in online and blended learning programs.
2. **Statistical techniques** such as Pearson correlation, simple linear regression, and multiple regression analysis were conducted using **SPSS** to test the proposed hypotheses.

Model Used:

1. The study adopted an **Emotion-Aware AI Conceptual Model**, integrating **Emotional Intelligence theory** with **Affective Computing systems** to assess student well-being.
2. The model positions **emotionally intelligent AI systems** as predictive tools influencing **academic engagement, emotional well-being, and dropout risk** in online classrooms.

Future Scope:

1. Future studies can adopt **longitudinal and experimental designs** to evaluate long-term psychological and academic outcomes of affective computing systems.
2. Further research may explore **cross-cultural comparisons, ethical AI frameworks, and advanced emotion-recognition models** in online education.

REFERENCE PAGE:

1. Danowitz, A., & Beddoes, K. (2018). *Characterizing mental health and wellness in students across engineering disciplines*. In **Proceedings of the American Society for Engineering Education Annual Conference**.
https://digitalcommons.calpoly.edu/cgi/viewcontent.cgi?article=1264&context=csse_fac
2. Reyes- Portillo, J. A., So, A., McAlister, K., Nicodemus, C., Golden, A., Jacobson, C., & Huberty, J. (2025). *Generative AI-powered mental wellness chatbot for college student mental wellness: Open trial*. **JMIR Formative Research**, 9, e71923.
<https://formative.jmir.org/2025/1/e71923/PDF>
3. *Synthesizing AI for mental wellness: Bridging artificial intelligence, machine learning, and personalized care*. (2024). **Journal of AI and Cognitive Science**, 9(9).
<https://media.neliti.com/media/publications/592672-synthesizing-ai-for-mental-wellness-and-d5445c69.pdf>
4. Hamdoun, S., Monteleone, R., Bookman, T., & Michael, K. (2023). *AI- based and digital mental health apps: Balancing need and risk*. **IEEE Technology and Society Magazine**, 42(1), 25–36.
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10063146>
5. Shende, H., Badode, R., Kar, R., Mohankar, P., & Sengupta, S. (2024). *AI being used to tackle mental health issues: Conversify—A robot*. **International Journal of Trend in Scientific Research and Development**, 8(5), 584–592.
<https://www.ijtsrd.com/other-scientific-research-a...>
6. Fontenelle-Tereshchuk, D. (2025). *First-year college students: Perspectives on technology and wellness in education*. **Trends in Higher Education**, 4, 13. <https://doi.org/10.3390/higheredu4010013>

7. Hanna, B. S. H., & Hanna, A. S. H. (2021). *AI-enabled smartphone-based intervention mental health application for university students*. Australasian Conference on Information Systems (ACIS) Proceedings, 89.

ACIS Proceedings reference page (find via AIS eLibrary):

<https://aisel.aisnet.org/acis2021/89/>

8. Shahzad, M. F., Xu, S., Lim, W. M., Yang, X., & Khan, Q. R. (2024). *Artificial intelligence and social media on academic performance and mental well-being: Student perceptions of positive impact in the age of smart learning*. **Heliyon**, **10**(8), e29523.

<https://doi.org/10.1016/j.heliyon.2024.e295239>.

9. Shifa, S., Hameed, S., Suhail, H., & Kiren, A. (2025). *Emotional intelligence meets artificial intelligence: Affective computing and student well-being in online classrooms*.

The Critical Review of Social Sciences Studies, **3**(3), 271–289.

This journal is less commonly indexed — title likely hosted on the journal's site (Critical Review of Social Sciences Studies).

Try:

<https://thecrsss.com/index.php/Journal/article/view/682>

10. Shifa, S., Hameed, S., Suhail, H., & Kiren, A. (2025). *Emotional intelligence meets artificial intelligence: Affective computing and student well-being in online classrooms*. **The Critical Review of Social Sciences Studies**, **3**(3), 271–289.

<https://thecrsss.com/index.php/Journal/article/view/271>

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